



UTM

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CREATIVITY AND INNOVATION - SSPG2113

TITLE: **TeaGel EcoFresh**

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1.0 INTRODUCTION

Tea is one of the most widely consumed non-alcoholic beverages in the world, with global production exceeding 6 million metric tons annually and projected to grow further as demand increases. This widespread consumption generates substantial volumes of tea waste, including spent tea leaves discarded after brewing as well as residues from pruning, processing, and packaging. Studies estimate that approximately 90% of the leaves become waste either during processing or post-brew, with these residues often disposed of in landfills or burned

The large quantity of waste presents both environmental and economic challenges. Improper disposal contributes to organic waste accumulation, potential soil and water contamination, and increased greenhouse gas emissions associated with biodegradation processes. At the same time, tea waste still contains valuable bioactive compounds such as polyphenols, tannins, and residual caffeine, which have functional properties including odor adsorption and insect deterrence.

Despite all this potential, the majority of tea waste remains under-utilized, with only a fraction repurposed for composting or industrial applications such as biochar production. The concept of converting waste into value-added products resonates with circular economy principles, which aim to reduce environmental impact and optimize resource efficiency. This project addresses the gap in sustainable reuse of tea waste by transforming spent tea leaves into a value-added solid gel air freshener with natural pest-repellent properties, thereby contributing to both waste reduction and the development of eco-friendly household products.

2.0 PROBLEM DEFINITION

The increasing consumption of tea has led to a significant rise in tea waste generation, most of which is disposed of without proper reuse. This contributes to environmental issues such as organic waste accumulation and greenhouse gas emissions. Simultaneously, many commercial air fresheners and pest repellent rely on synthetic chemicals that may pose health risks and negatively affect indoor air quality.

There is a lack of sustainable, low-cost, and eco-friendly household products that effectively utilize organic waste materials while maintaining functional performance. This highlights the need for innovative solutions that address both waste management and consumer health concerns.

2.1 OBJECTIVE

The objectives of this project are as follows:

1. To utilize used tea waste as the primary raw material in the production of a solid gel air freshener.
2. To develop a solid gel product that functions as both an air freshener and a natural pest repellent.
3. To reduce environmental impact by promoting the reuse of organic waste materials.
4. To provide a safer and more sustainable alternative to conventional chemical-based air fresheners.
5. To evaluate the feasibility of tea waste as a value-added commercial product.

3.0 CONCEPTUAL PROTOTYPE

3.1 Product Design & Mechanism

TeaGel EcoFresh is a sustainable dual-action solid gel designed to bridge the gap between home fragrance and pest control. The prototype focuses on "Upcycled Innovation," transforming organic waste into a functional household utility.

Core Components

- Upcycled Tea Waste: Extracts bioactive compounds (Caffeine & Polyphenols) from used tea leaves to serve as the active repellent agent.
- Agar/Gelatin Matrix: A biodegradable base that facilitates a slow-release mechanism, ensuring the fragrance and repellent vapors are dispersed consistently over time.
- Odor Neutralizers: Natural elements that eliminate unpleasant smells rather than just masking them with heavy perfumes.

4.0 SCAMPER ANALYSIS

Table 2: SCAMPER Analysis of TeaGel EcoFresh

Letter	Dimension	Application in TeaGel EcoFresh
S	Substitute	Instead of relying on inconsistent small-scale collection from households, the supply chain is substituted by partnering with centralized industrial sources, such as hotel chains or tea factories. This ensures a consistent, high-volume tonnage of tea waste, which stabilizes production schedules and standardizes the quality of the raw material.

C	Combine	The product combines the tea waste extract (containing caffeine and polyphenols) with other natural compellents (e.g., citronella or lemongrass). This combination creates a synergistic effect, boosting the product’s overall effectiveness against a wider variety of specific household pests while maintaining a pleasant, layered fragrance profile.
A	Adapt	The formulation is adapted to create specialized versions for different rooms. For example, a “Kitchen” version could be adapted to specially neutralize strong food odors (like fish or garlic), while a “Bathroom” version could focus on fresh scents to mask dampness. This makes the product versatile across the entire household.
M	Modify	The product is modified by increasing the ratio of tea waste extract within the gel matrix. By creating a more concentrated formula, the emission of repellent vapors is intensified. This effectively modifies the “repellent radius”, allowing a single jar to protect a larger square footage than the standard formulation.
P	Put to another use	The product is designed to be put to another use at the end of its life cycle. Once the gel dried out and stopped releasing fragrance, the residue –which is composed of biodegradable plant-based agar/gelatin and organic tea leaves, can be removed from the container and used as nutrient-rich plant fertilizer, ensuring a zero-waste lifecycle.

5.0 INNOVATION CATEGORIZATION

Table 1: Innovation Categorization of TeaGel EcoFresh

Dimension	Choice	Justification
Focus	Product	Teagel Eco Fresh is new because of the gel itself. It combines two functions: removing bad smells and keeping pests away. The change is in the product, not in the way it is made.
Magnitude	Radical	It is a big change because it uses tea waste in a new way. This needs new knowledge about natural ingredients and sustainability, not just small improvements to old products.
Impact	Competence-destroying	It challenges companies that make chemical air fresheners and insect sprays. Their old skills with synthetic chemicals may not be useful anymore if people prefer eco-friendly options.

6.0 SWOT ANALYSIS/ADVANTAGES

Table 2 :SWOT analysis and advantages of TeaGel EcoFresh

Category	Factors	Advantages
Strength (S)	- Eco-friendly and sustainable (upcycled tea waste) - Dual functionality (air freshener & pest repellent)	- Reduces environmental impact - Provides higher value compared to single-purpose

	• Natural ingredients	products • Safe for children, pets, and sensitive users
Weakness (W)	• Limited market awareness • Shorter shelf life concerns • Higher production cost	• Opportunity to build strong branding and product education • Reinforces natural and chemical-free product positioning • Supports premium pricing and perceived product quality
Opportunity (O)	• Growing Eco-conscious market • Partnerships with tea shops • Educational marketing	• Strong market alignment and high acceptance potential with sustainability trends • Reduce raw material cost while creating economic value from waste • Builds credibility and enhances adoption through knowledge-based promotion

		Risks
Threats (T)	• Competition from established brands • Market price sensitivity • Environmental factors	• Consumer preference for familiar brands • Perceived value must be clearly communicated • Inconsistent user experience

7.0 STRATEGIES PLANNING ROADMAP

Problems/Troubles

- Mild pest repellency compared to chemical products
- Limited diffusion speed in large or well-ventilated rooms
- Gel effectiveness depends on placement location
- Evaporation rate increases in hot environments
- Competition from established brand

Improvement Strategies

- Optimize tea extract concentration to enhance effectiveness
- Improve gel formulation for longer-lasting release
- Combine tea waste with selected natural essential oils
- Design containers with adjustable airflow vents
- Develop an optional vapor-based or aroma-diffusion form, allowing the active compounds to disperse faster into the surrounding atmosphere

- Explore low-heat or passive diffusion methods to improve spread without combustion

Future Potential

- Introduce gel, spray, and vapor diffuser variants
- Adapt the technology for larger indoor spaces
- Integrate with electric or passive diffuser devices
- Expand applications to hotels, offices, restaurants and public areas
- Apply the same concept to other agricultural waste materials

The roadmap can be divided in 3 stages that are Problems, Improvement and Future Potential. We consider it based on what will happen in the future, the challenges producing this product and how it can survive in world competition.

At the first stage (Problems/Troubles), TeaGel faces several limitations that affect its overall performance. The pest-repellent effect is relatively mild compared to chemical-based products, and the release of active compounds can be slow, especially in large or well-ventilated spaces. In addition, the gel's effectiveness may decrease in hot environments due to faster evaporation. These challenges indicate the need for further improvement to enhance diffusion efficiency and product lifespan.

The second stage is Improvement Strategies. To address these challenges, several improvement strategies can be explored. The formulation of TeaGel can be optimized by increasing the concentration of tea extract and improving the gel matrix to extend its longevity. Natural essential oils may also be added to enhance fragrance and repellency. Furthermore, TeaGel can be developed into an optional vapor-based or aroma diffusion form, allowing the active compounds to spread more quickly and evenly in the surrounding atmosphere without combustion. Improved container design with controlled airflow can further enhance performance.

Finally is Future Potential. With continuous improvement, TeaGel has strong potential to expand beyond its current solid gel form. Future developments may include multiple product variants such as solid gel, vapor diffuser, or refillable systems for larger indoor spaces. The concept can also be scaled for commercial applications in homes, offices, and public areas. Additionally, the same upcycling approach may be applied to other types of agricultural waste, supporting long-term sustainability and broader environmental impact.